11.9 The Safety Parabola

WORKSHEET

The question

STEP II 2014, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q10

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Write x and y in terms of t, and eliminate t. How can you write $\sec^2\theta$ in terms of $\lambda = \tan\theta$?

STAGE 1 Fix x, vary the angle

Covers: the maximum height Y above a given x

Pause and try Fix $x > 0$ and think of y as a function of λ alone. What kind of function is it? Find its greatest value. (Or: when does the quadratic in λ have a real root?)

STAGE 2 The region

Covers: the graph and the reachable region

Ask yourself Sketch the curve Y: where is its vertex, and where does it meet the ground? Which points can be reached? How many trajectories pass through a point below the curve, and through a point on it?

STAGE 3 The furthest point

Covers: the furthest distance from O

Ask yourself The furthest point lies on the boundary. Write $x^2 + Y^2$ as a function of $X = x^2$. Is it increasing or decreasing? Where along the boundary is it largest?

What's really going on

Envelopes

A one-parameter family of curves $\mathrm{F}(x, y, \lambda) = 0$ often has an *envelope*: a curve touching every member of the family. It is found by solving $\mathrm{F} = 0$ and $\frac{\partial \mathrm{F}}{\partial \lambda} = 0$ together, which is exactly what we did by maximising y over λ. Envelopes appear throughout science. The bright curve of light on the surface of a cup of coffee is the envelope of reflected rays, a *caustic*. A supersonic aircraft's sonic boom is carried by the envelope of the sound waves it emits, the *Mach cone*. And in Section 8.1, the envelope of a family of solutions of a differential equation was itself a solution, a "singular" one.

The focus

The parabola of safety has a remarkable property: its focus is the launch point O. (A parabola $y = c - \frac{x^2}{4c}$ has its focus at the origin; here $c = \frac{u^2}{2g}$.) Every trajectory, too, is a parabola, and the foci of all the trajectories lie on a circle of radius $\frac{u^2}{2g}$ about O. There is an energy explanation: the speed of the projectile at height y is $\sqrt{u^2 - 2gy}$, which is zero at the height $\frac{u^2}{2g}$ of the directrix, and parabolas whose directrix is the same line and which pass through O have their foci on that circle.

Torricelli's parabola

Galileo showed around 1609 that projectiles move along parabolas, and that the maximum range on level ground is at 45°. His student Evangelista Torricelli, better known for inventing the barometer, found the parabola of safety in 1644, in his *De motu gravium*. For gunners it answered a practical question: which targets are within range at all? Torricelli also showed that the trajectories through a given point come in

pairs with angles symmetric about the direction of the point and the vertical, the high and low shots of the question.

Try this next

1 **An exploding firework.** Try STEP II 2023, Question 10 (in OCR's download of the 2023 papers, www.ocr.org.uk/Images/707102-2023-past-papers.zip). It views the particles from an explosion as a sphere whose centre falls freely, and finds a particle that runs along the paraboloid of safety.

2 **Firing up a slope.** Find the maximum range of a projectile with speed u up a plane inclined at angle β. (*Hint:* where does the line $y = x\tan\beta$ meet the parabola of safety?) Show that the best angle bisects the angle between the slope and the vertical.

3 **Two angles.** Show that the two angles that reach a point P are symmetric about the bisector of the angle between OP and the vertical.

4 **The foci.** Show that the trajectory with angle θ has its focus at $\left(\frac{u^2}{2g}\sin 2\theta, -\frac{u^2}{2g}\cos 2\theta\right)$, and hence that the foci lie on a circle about O.

Further reading

- **David Morin, *Introduction to Classical Mechanics*** (Cambridge University Press), Chapter 3, which discusses the envelope of trajectories.
- **J. W. Bruce, P. J. Giblin & C. G. Gibson, "On caustics of plane curves"**, *American Mathematical Monthly* 88 (1981). Envelopes and caustics in geometry.
- **E. H. Lockwood, *A Book of Curves*** (Cambridge University Press). Envelopes, caustics and many classical curves, with constructions.

The full guided solution is in Section 11.9:
https://beyondstep.keithmcnulty.org/ch11/safety-parabola.html